8 The perimeter of a right-angled triangle is 72 cm.
The lengths of its sides are in the ratio 3 : 4 : 5

Work out the area of the triangle.

3 + 4 + 5 = 12 equal ratio units

One unit = 72 ÷ 12 = 6 cm

1 mark
Divide the perimeter by 12

A, B and C are the three side lengths, in cm.

A		B		C	
3	×6	4	×6	5	×6
18		24		30	

1 mark
Find sides 18 cm, 24 cm and 30 cm

30 cm is the hypotenuse. The perpendicular sides are 18 and 24.

Area = ½ × 18 × 24 = 9 × 24

1 mark
Use half the base times height

= 9 × 20 + 9 × 4 = 180 + 36 = 216 cm²

1 mark
Correct triangle area

216 cm²

(Total for Question 8 is 4 marks)